MRV Best Practices Guide

Monitoring, Reporting, and Verification for Land-Based Carbon Projects

Reading 5 · Module 4 · Meer Foundation EduSuTo

Course: Climate Change: Carbon Capture and Storage

What Is MRV and Why Does It Matter?

Monitoring, Reporting, and Verification (MRV) is the backbone of any carbon project. Without robust MRV, a carbon credit is just a number on paper. With rigorous MRV, it is a verified, traceable, internationally recognised unit of climate action that can be traded, retired, and counted toward net-zero commitments.

The three components are distinct:

- **Monitoring:** The systematic, ongoing collection of data needed to quantify project outcomes. Happens throughout the project lifetime.
- **Reporting:** The organisation, documentation, and submission of monitoring data in the format required by the applicable carbon standard.
- **Verification:** Independent, third-party assessment that the monitoring data and reported outcomes are accurate, complete, and consistent with the approved methodology.

Part 1: Monitoring Design Principles

Good monitoring design follows six core principles:

Principle	Description	Application to Soil Carbon
Relevance	Monitor what matters for the outcome being claimed	SOC must be measured, not proxied alone
Completeness	Cover all emission sources and sinks in the project boundary	Soil, above-ground biomass, and farm inputs
Consistency	Use same methods each monitoring period for comparability	Same lab, same season, same depth each year
Accuracy	Minimise measurement uncertainty	Stratified sampling; sufficient sample density

Principle	Description	Application to Soil Carbon
Transparency	Document methods fully so results are reproducible	Full sampling records, chain of custody for samples
Conservativeness	Where uncertain, underestimate, not overestimate	Apply uncertainty deductions to credit claims

Part 2: Field Data Collection — Best Practices

Soil Sampling Protocol:

- Define strata based on soil type, land use history, and topography before sampling
- Use a stratified random sampling design — not convenience sampling
- Minimum 15 composite cores per stratum; each composite from 5+ sub-samples
- Record GPS coordinates (±5m accuracy) for every sampling point
- Collect bulk density rings (undisturbed cores) at 3–5 points per stratum
- Label samples with project ID, stratum, date, collector name
- Transport in sealed, labelled plastic bags; refrigerate within 24 hours of collection

Activity Data Collection:

- Record all management practices with dates (planting, harvest, tillage, input application)
- Photograph each field at each monitoring event (GPS-tagged photos preferred)
- Maintain input purchase records (seeds, fertiliser, compost) as supporting evidence
- Collect farmer signatures on monitoring forms as proof of participation

Part 3: Digital MRV (DMRV) Tools

Tool Type	Examples	Application	Cost Indicator
Mobile data capture	ODK Collect, KoboToolbox, Survey123	Field data collection with GPS and photos	Free / Low
Satellite imagery (optical)	Sentinel-2, Landsat, Planet	Land cover change, NDVI monitoring, crop mapping	Free (Sentinel/Landsat) / Paid (Planet)
Satellite radar (SAR)	Sentinel-1	Above-ground biomass, soil moisture	Free

Tool Type	Examples	Application	Cost Indicator
Spectroscopy (NIR/MIR)	Portable spectrometers	Rapid SOC estimation in field	High capital; low per-sample
Blockchain registry	Toucan, Regen Network, Gold Standard Digital	Credit issuance, retirement, and traceability	Variable
AI/ML models	Custom models trained on lab+satellite data	SOC estimation at scale from satellite data	Requires calibration
Remote sensing platforms	Google Earth Engine, EO Browser	Data processing and analysis	Free / Low

Part 4: Verification — What Auditors Look For

Third-party verification is conducted by accredited Validation and Verification Bodies (VVBs). The verification process typically covers:

Documentation review: Does the PDD accurately describe the project? Are all required sections complete? Do the methodology requirements match the implementation?

Site visits: Physical inspection of a sample of project sites. Farmer interviews. Verification that practices described in the PDD are actually happening.

Data audit: Statistical audit of monitoring data. Checking for outliers, data entry errors, implausible values. Laboratory certification verification.

Calculation verification: Recalculating credit estimates from raw data. Checking that formulas match the approved methodology.

Additionality assessment: Confirming that the project activity is genuinely additional — it would not have occurred under the baseline scenario.

Common Verification Findings That Reduce Credits: • Insufficient sample density for claimed area • Missing bulk density measurements • Samples collected outside the approved sampling window • Activity data gaps (farmer records incomplete) • Practices not applied to full claimed area • Laboratory using non-approved methods